React Training — Day 3

Components & Reusable UI

🎯 Goal of Today

Understand

Learn exactly what React components are and why they are essential.

Create

Learn how to create your own functional components from scratch.

Reuse

Discover how to reuse components to build complex UI efficiently.

Organize

Learn best practices for structuring files in a React project.

What is a Component?

A component is a **reusable piece of UI**.

Instead of writing one large webpage, React breaks the UI down into small, isolated, and reusable parts.

Examples:

- ▹ Navbar
- ▹ Sidebar
- ▹ Video Player
- ▹ Comments Section

Each of these independent parts is a React component.

Real-Life Example

Think of a standard YouTube page structure.

- **Header:** Search bar, profile (Navbar)
- **Left Menu:** Subscriptions (Sidebar)
- **Main Area:** The Video Player
- **Right Panel:** Recommended Videos
- **Bottom Area:** Comments

Each distinct section is built as an individual component. Companies like YouTube, Netflix, and Instagram use this exact architecture to scale their platforms.

Types of Components

1️⃣ Functional Components ✅

This is the modern way to write React.

They are simply JavaScript functions that return HTML (JSX). They are easier to read, test, and form the backbone of modern React applications using Hooks.

Today we focus exclusively on Functional Components.

2️⃣ Class Components ❌

This is the old, legacy way to write React.

They use ES6 Javascript Classes. While you may still see them in older codebases, they are no longer recommended for new projects.

First React Component & Usage

Defining a Component

A component is just a function that returns JSX.

```
function Navbar() { return My Website }
```

This creates a reusable component called `Navbar`.

Using a Component

Components behave exactly like custom HTML tags.

```
function App() { return (    ) }
```

React sees and renders the UI defined inside it.

Professional Project Structure

As your project grows, you cannot keep all components in one file.

- Create a dedicated **components** folder inside `src`.
- Create a new file for every single component.
- Capitalize component filenames (e.g., `Navbar.jsx`).
- Keep `App.jsx` as the main parent component.

Keeping components organized is critical in real-world projects.

```
📂 src ├ 📂 components │ └ 📄 Navbar.jsx ├ 📄 App.jsx └ 📄 main.jsx
```

Create & Import Component

1. Create: 📄 components/Navbar.jsx

Define the component and export it.

```
function Navbar() { return ( My Website Navbar )
} export default Navbar
```

2. Import: 📄 App.jsx

Import the file to use it in App.

```
import Navbar from "./components/Navbar"
function App() { return (   Welcome to my
website  ) } export default App
```

Building Multiple Components

Expanding the App

You can create as many components as you need to build out your layout.

```
📂 components
├ 📄 Navbar.jsx
├ 📄 Footer.jsx
└ 📄 ProductCard.jsx
```

This modularity is how large React projects are structured.

Create: 📄 components/Footer.jsx

Let's add a simple Footer component.

```
function Footer() { return ( Copyright 2026 ) } export default Footer
```

Use Multiple Components

Composing the UI

Bring all your pieces together in the main `App.jsx` file.

By importing both the **Navbar** and the **Footer**, you start to form a complete webpage structure simply by stacking custom HTML-like tags.

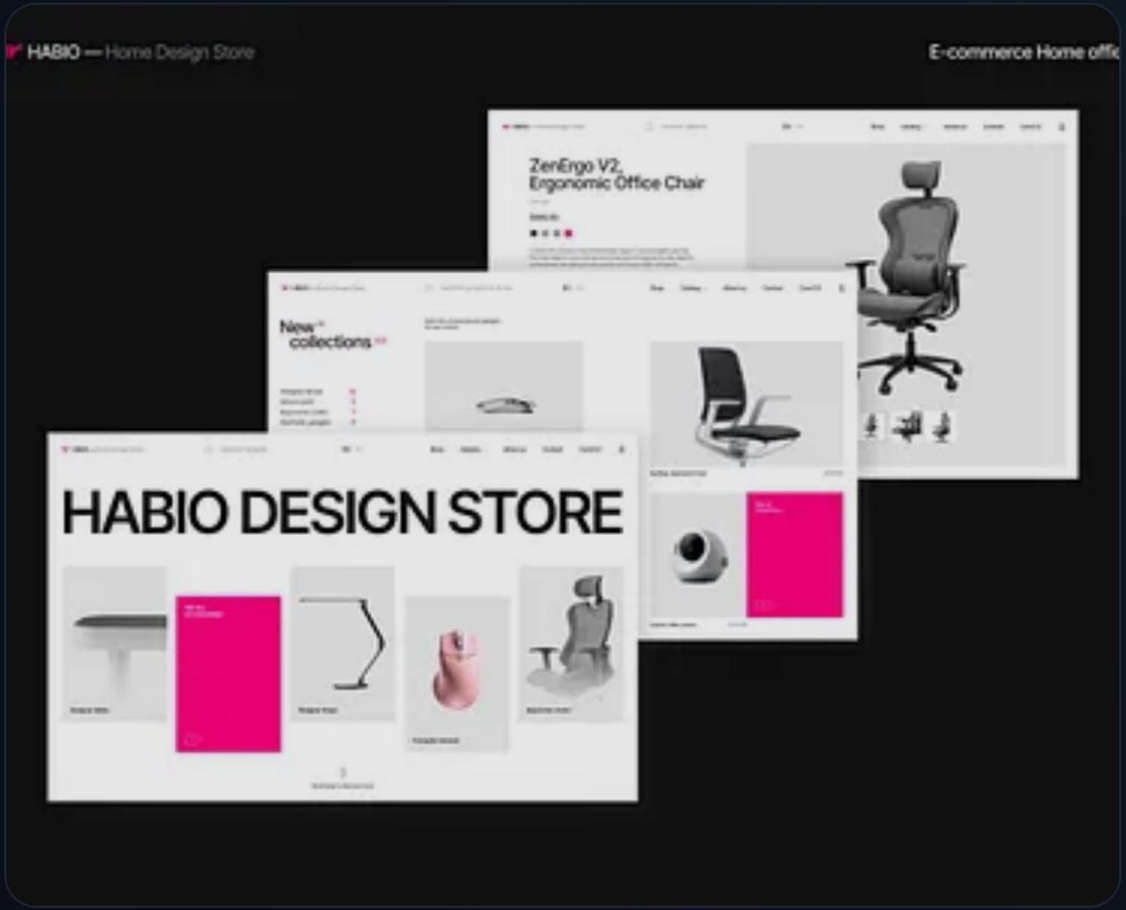
📄 `App.jsx`

```
import Navbar from "./components/Navbar" import
Footer from "./components/Footer" function App()
{ return (   Homepage   ) }
```

Reusable Components

This is the **real power** of React.

Consider an E-commerce website. Instead of writing the complex HTML for a product UI repeatedly, you write it once.



One component → many uses.

Product Card Component

1. Create: 📄 ProductCard.jsx

```
function ProductCard() { return (
className="card"> iPhone Price: ₹70000 Buy Now
) } export default ProductCard
```

2. Reuse in: 📄 App.jsx

```
import ProductCard from
"./components/ProductCard" function App() {
return (      ) }
```

The exact same UI appears multiple times effortlessly.

Practice Tasks

Time to apply what you've learned.

Your Assignments

1

Task 1: Setup

Create three distinct components in your components folder:

- ▷ `Navbar.jsx`
- ▷ `Profile.jsx`
- ▷ `Footer.jsx`

Import and add them all to `App.jsx`.

2

Task 2: UI Design

Build out the **Profile Component** UI to display:

```
Name: Saurav Role: React Developer Goal: Full Stack Engineer
```

3

Task 3: Reuse

Prove the reusability of your code.

In `App.jsx`, display **3 profile cards** using the exact same component.

What You Learned Today

- **What components are:** Independent, reusable pieces of UI.
- **Functional components:** The modern standard using standard JS functions.
- **Creating files:** Isolating UI logic into individual `.jsx` files.
- **Import & Export:** Connecting different files using ES6 syntax.
- **Reusable UI:** Rendering the same component multiple times cleanly.
- **Professional Structure:** Using a `src/components` architecture.

Tomorrow (Day 4)

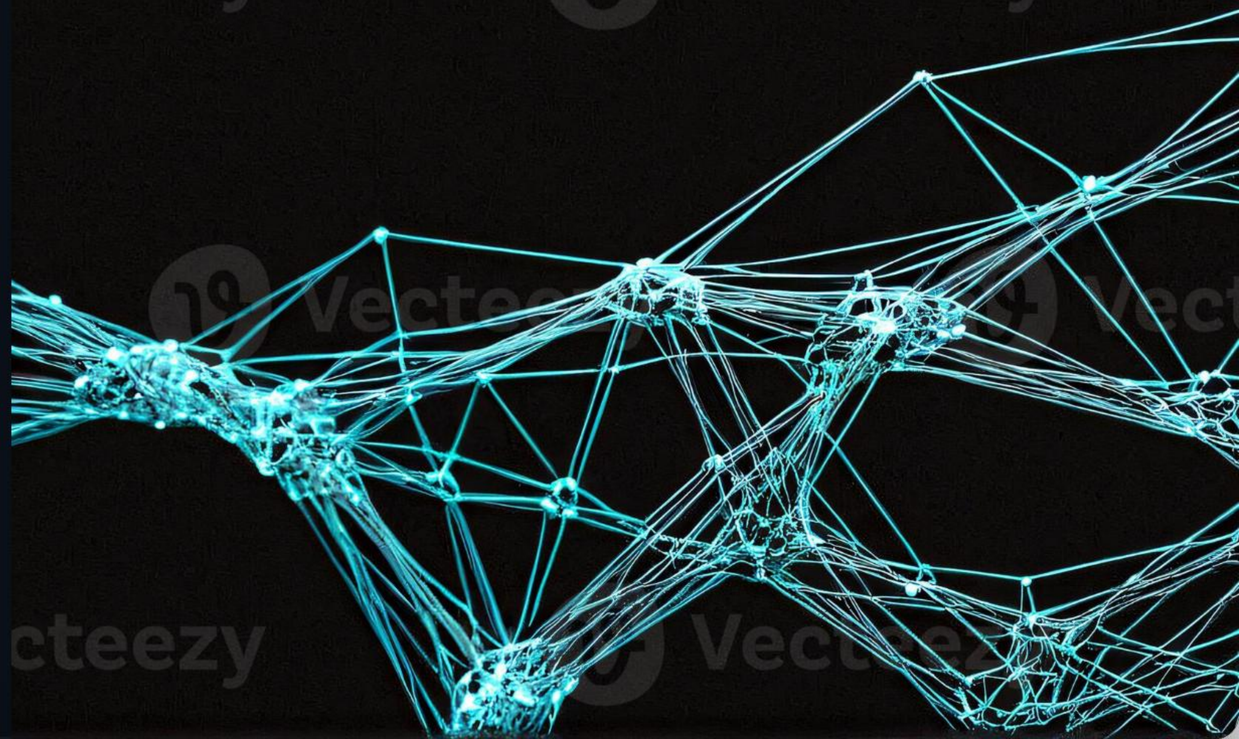
Next Topic: 🔥 Props

Props (Properties) are **very important**. They allow components to receive unique data.

```
 name="iPhone" price="70000"/>  name="Samsung"
price="50000"/>
```

Same Component → Different Data

You will learn to build dynamic E-commerce UI cards.



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